



Gaston Planté battery, 19th century

SAFER TO HANDLE

Early batteries used liquids, often contained in glass, and had to be moved very carefully.

With the invention of his dry battery in 1886, the German scientist Carl Gassner made batteries easier to handle and usable with more devices. His battery used a dry paste electrolyte (the substance through which an electric current passes) enclosed in a zinc case.



One of Hornsdale Wind Farm's 99 wind turbines

STORING UP ENERGY

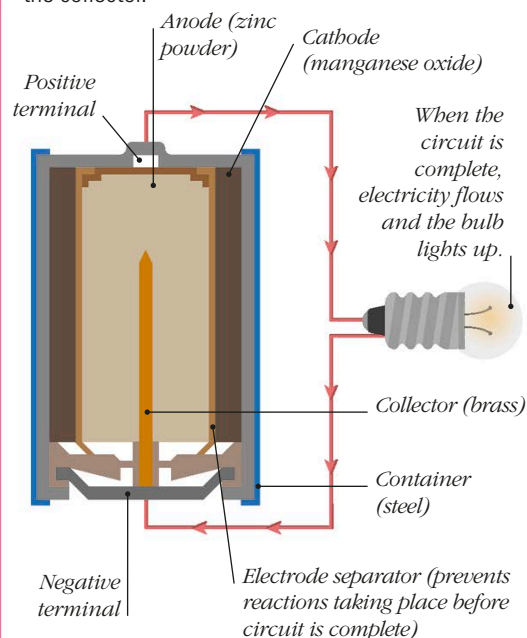
Batteries not only provide electricity, they can also be used to store it. Many companies now produce batteries that store electricity gathered from solar panels or wind farms. This electricity can be used to provide energy at times when there is no sunshine or no wind blowing, either locally or to the national electricity supply.

LONGER LIFE

Early Voltaic-style batteries did not last very long. They ran on chemicals and once their supply had run out, they stopped working. The French physicist Gaston Planté overcame this problem with his invention of a rechargeable lead-acid battery in 1859. Many modern batteries, such as the lithium-ion batteries in smartphones, are rechargeable.

HOW A BATTERY WORKS

When an alkaline dry battery is placed in a device, it causes a chemical reaction to take place between the battery's two electrodes (the anode and the cathode), producing electricity. The electricity is then conducted out of the battery to the device via the collector.



◀ LARGEST BATTERY

The world's biggest storage battery was built for the Hornsdale Wind Farm, Australia, in December 2017, by the American company Tesla. This giant array of lithium-ion batteries can store enough electricity to power 30,000 homes.

